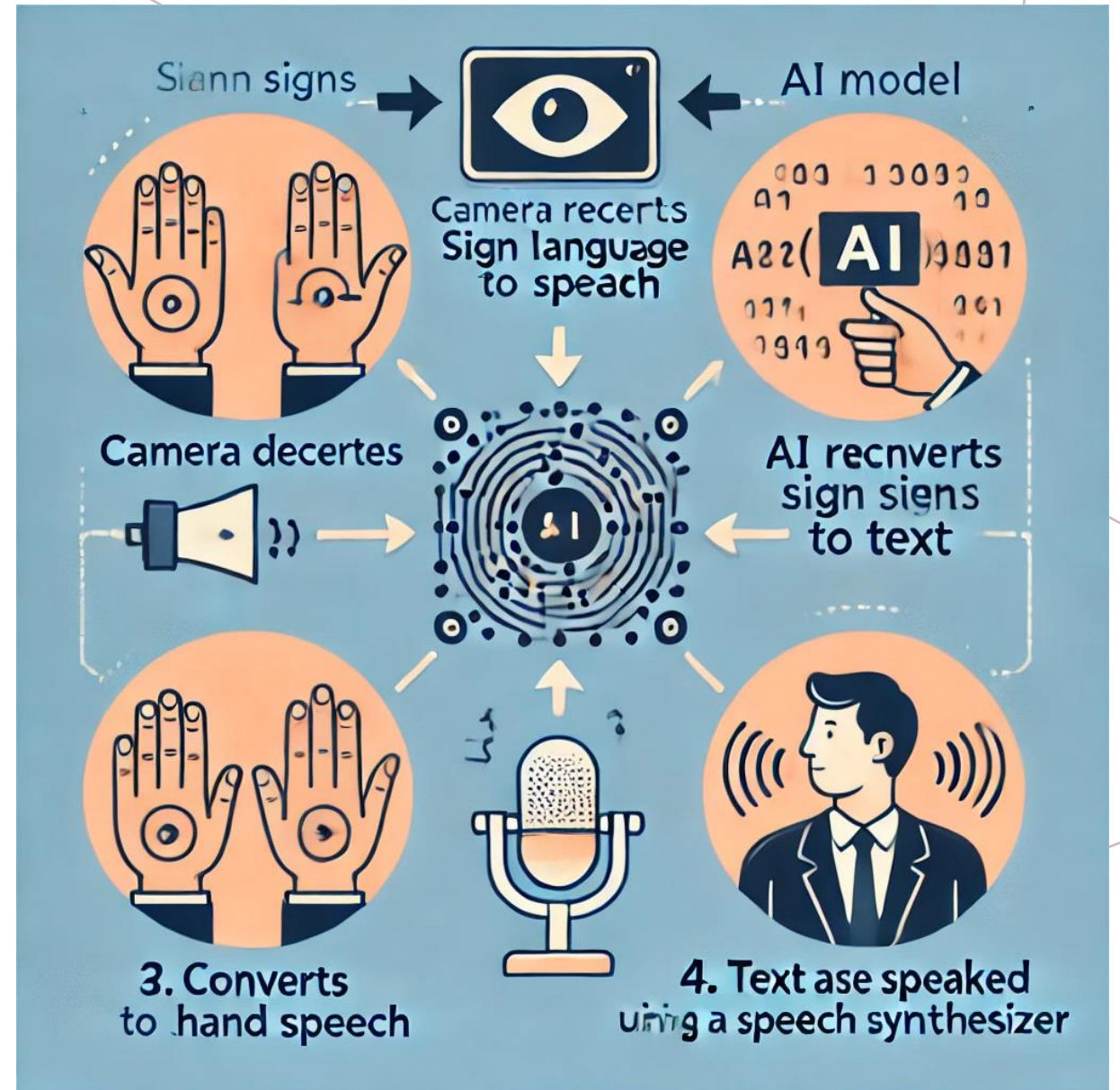


MEOW

"BREAKING BARRIERS WITH AI-POWERED SIGN LANGUAGE RECOGNITION"



TEAM - AKATSUKI



JAYDEV
SEMWAL



ANANT
GARG



TANUJ
RAJPUT

PROBLEM STATEMENT

- Over **466 million** people worldwide have hearing or speech impairments.
- Communication barriers make daily interactions challenging.
- Existing solutions are either expensive or not widely accessible.
- How can technology bridge this gap?





SOLUTION - MEOW

- **MEOW** is an AI-powered Python software that:
- Captures **sign language gestures** via webcam.
- Recognizes and translates them into **text**.
- Converts the text into **speech**, allowing seamless communication.
- **Affordable, accessible, and real-time.**

HOW IT WORKS?

CAPTURING GESTURES

USES OPENCV TO ACCESS WEBCAM INPUT.

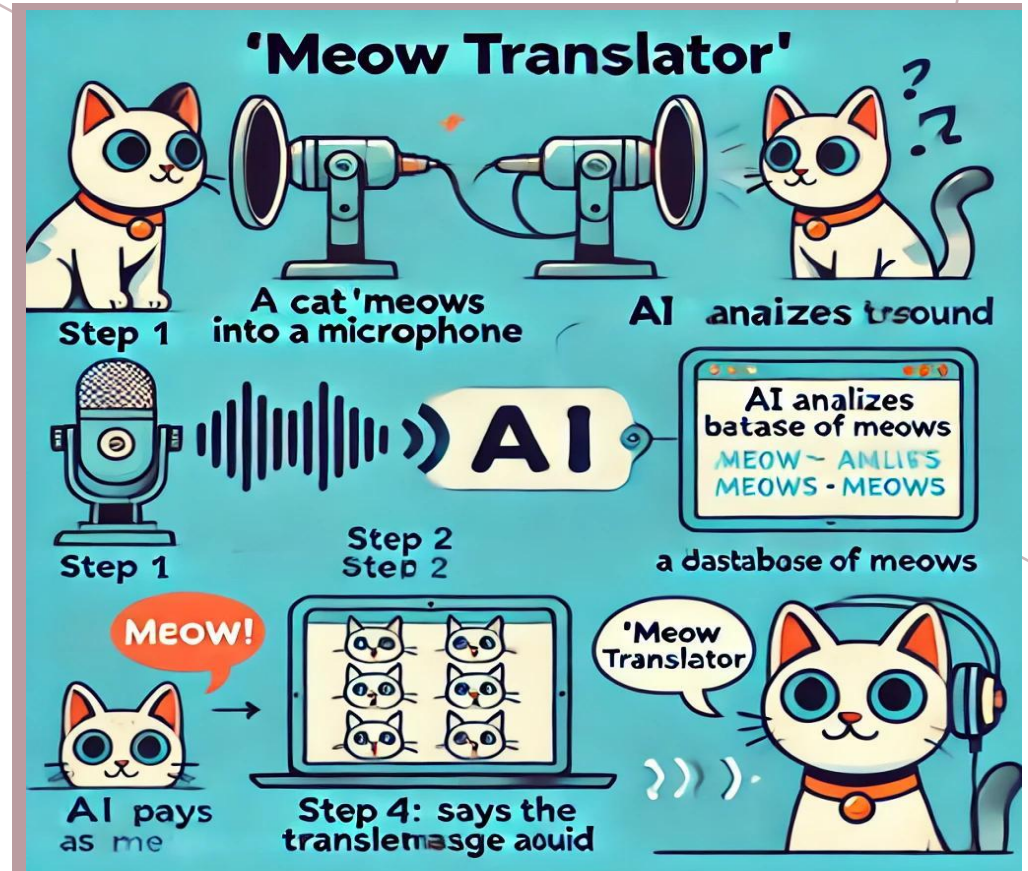
RECOGNIZING SIGNS

USES MEDAPIPE HANDS FOR REAL-TIME HAND TRACKING.

ANALYZES FINGER POSITIONS TO IDENTIFY SIGN LANGUAGE GESTURES.

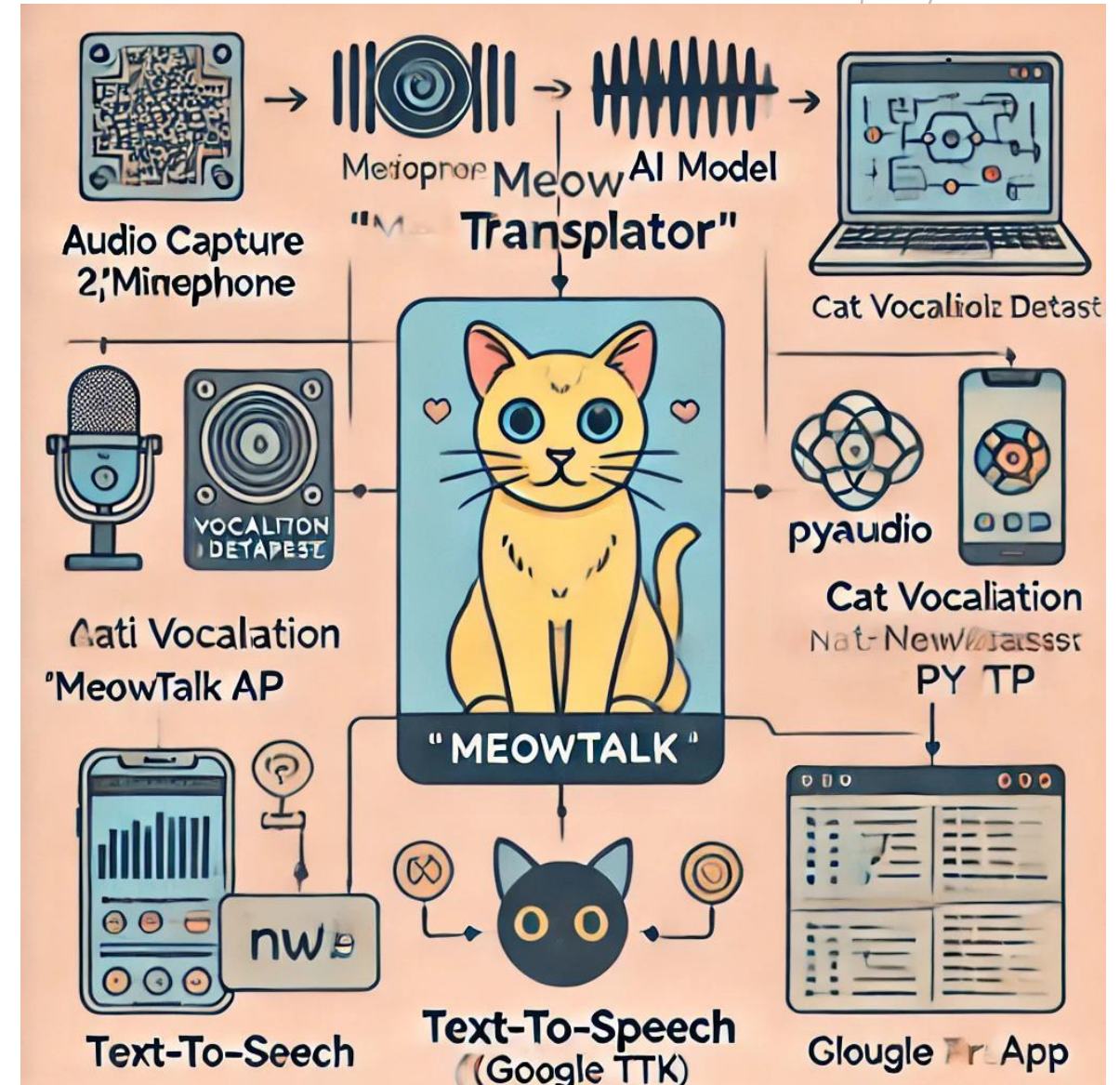
CONVERTING TO SPEECH

USES PYTTX3 (TEXT-TO-SPEECH) TO VOCALIZE THE DETECTED GESTURES.



TECH STACK

- Programming Language: Python
- Libraries Used:
 - OpenCV - Webcam integration & frame processing
 - MediaPipe - Hand tracking & gesture recognition
 - pyttsx3 - Text-to-Speech synthesis



FUTURE SCOPE

- Expand to **ASL (American Sign Language) & ISL (Indian Sign Language)**.
- Add **ML-based models** for advanced recognition.
- Improve UI/UX for a **more user-friendly experience**.
- Deploy as a **mobile app** for better accessibility.

CLOSING STATEMENT

- "MEOW is more than software—it's a step towards an inclusive world."
- "With AI, we are giving a voice to those who need it the most."
- Thank You! Any Questions?



*THANK
YOU*

